

Final Project Report

Research Submission and Evaluation System for University



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Submitted By

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CERTIFICATE

This is to certify that Khadeeja Kamran (BC220203838) has worked on and completed her Software Project at Software & Research Projects Section, Department of Computer Sciences, Virtual University of Pakistan in partial fulfillment of the requirement for the degree of BS in Computer Sciences under my guidance and supervision.

In our opinion, it is satisfactory and up to the mark and therefore fulfills the requirements of BS in Computer Sciences.

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EXORDIUM

In the name of Allah, the Compassionate, the Merciful.

**Praise be to Allah, Lord of Creation,
The Compassionate, the Merciful,
King of Judgment-day!**

**You alone we worship, and to You alone we pray for help,
Guide us to the straight path**

The path of those who You have favored,

**Not of those who have incurred Your wrath,
Nor of those who have gone astray.**

DEDICATION

This project is dedicated to my younger self, who once doubted her abilities and wondered if she was truly capable of programming. To her, this work stands as proof that persistence can transform uncertainty into achievement.

It is also dedicated to my future self — the graduate and professional I am becoming — who will build upon this foundation and continue striving for greater milestones.

Finally, I dedicate this accomplishment to my parents and siblings. My father's unwavering belief in me, my mother's gentle concern for my progress, and the ever-present support of my sister have been the strength behind this journey. Their faith and encouragement formed the unseen pillars that carried me to this stage.

ACKNOWLEDGEMENT

Foremost in my gratitude is Allah (SWT), the Most Merciful and the Granter of success. Without His guidance and countless blessings, I would not have reached the stage of completing this final report.

I am sincerely grateful to my university for providing the platform to undertake this project, and to my supervisor for their valuable role in evaluating my submissions, presenting assignment descriptions in a clear, accessible manner, and offering meaningful feedback.

This was a challenge I would not have pursued voluntarily, yet as a requirement it pushed me beyond my comfort zone and helped me achieve more than I expected.

Developing my first complete website was a milestone I once thought beyond my reach, yet with this support it became a reality.

I would also like to acknowledge the valuable tutorials and guidance shared by various YouTubers, whose content greatly assisted me in understanding complex concepts and implementing solutions during this project.

PREFACE

This report has been prepared as a fulfillment requirement for the degree of Bachelor of Computer Science (BSCS) for the academic session 2022–2026.

The project, titled “Research Submission and Evaluation System”, lies within the domain of Web Programming and is designed to streamline the submission, evaluation, and feedback process for university research papers. It enables seamless student submissions, efficient faculty evaluations, and transparent communication, while administrators benefit from improved oversight and a centralized repository of published work. The system ultimately fosters a collaborative and accountable academic environment.

This report encompasses the complete journey of the project—from the **Software Requirements Specification**, through the **Document Design stage**, and culminating in the highlights of its construction. It reflects both the technical development and the vision of creating a platform that unifies students, faculty, and administrators in a structured academic ecosystem.

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CHAPTER 1

Gathering & Analyzing Info

1.1 Introduction

“The Research Submission and Evaluation System” is a web-based platform developed to streamline the submission, evaluation, and feedback process of university research papers. It enables students to submit their work seamlessly, while faculty members and supervisors can efficiently evaluate and provide constructive feedback.

By eliminating the inefficiencies of traditional paper-based submissions, the system enhances tracking, ensures transparency, and fosters effective communication between students and evaluators. Administrators also gain the ability to monitor and manage submissions and reports with ease, while the centralized repository of accepted and published research provides a reliable archive for academic recognition.

1.2 Purpose

The aim of the system is to create a transparent, collaborative, and efficient academic environment for research evaluation. It ensures that students can upload their work in PDF or MS Word format, receive timely evaluations, and benefit from constructive feedback. At the same time, faculty and administrators can manage submissions more effectively, improving oversight and accountability. Ultimately, the system supports academic growth by simplifying workflows, strengthening communication, and maintaining a structured repository of published research.

The purpose of this chapter is to provide a clear, structured, and detailed description of requirements before the start of system development.

1.3 Scope

The scope of “The Research Submission and Evaluation System” outlines what the system will provide to students, faculty, and administrators, ensuring clarity on its coverage and limitations. The system,

- **Streamlines Research Submission** – Students can submit research papers in PDF or MS Word format along with relevant metadata, ensuring a structured submission process.
- **Facilitates Evaluation & Feedback** – Faculty members/supervisors assess research papers based on predefined criteria, provide feedback, and upload commented files or ask for revisions if necessary.
- **Enables Notifications & Communication** – The system sends email notifications regarding submission status, evaluation results, and other actions to keep students and faculty informed.
- **Enhances Transparency & Tracking** – Students can track the status of their submissions, receive evaluation results, and update research papers while maintaining a version history.
- **Generates Report** – Admin and faculty members/supervisors can generate reports on submission statistics, evaluation progress, and list of published papers.
- **Provides a Central Repository** – Accepted and published research papers are stored in a searchable centralized database, accessible to all authorized users.

1.4 Definitions, acronyms and abbreviations

Definitions

- **Submission** – The process by which students upload their research papers to the system.
- **Evaluation** – The review of submitted research papers by faculty members/supervisors based on predefined criteria.
- **Feedback** – Constructive comments, suggestions, or revision requests provided by evaluators to students.
- **Repository** – The centralized database where accepted and published research papers are stored.
- **Administrator** – A user role responsible for managing submissions, reports, and overall system oversight.

Acronyms

- **RSES** – Research Submission and Evaluation System

1.5 Project requirements

1.5.1 Functional Requirements

FR-1: User Registration and Authentication

- Students must be able to create an account using their university email address and personal details including, name and study field.
- Faculty members/ supervisors must be able to register a new account by providing the necessary details.

FR-2: Role-based Access Control

Different roles come with different levels of access

- **Students:** Can submit research papers, track status, view evaluations/feedback, download commented files.
- **Faculty members:** Can create research assignments, review/evaluate submissions, provide feedback, rate research papers, and publish accepted papers.
- **Admin:** Can manage user roles (accept / reject user registration requests), monitor submissions, tracking, and generate reports.

FR-3: Research Paper Submission

- Students can submit research papers (in PDF or MS Word format only) against the relevant research assignment.
- The system should provide interactive interface to students to provide metadata (e.g., Student ID, Study Program Enrolled, Paper Title, Abstract, keywords and Category / Domain).
- The system should allow students to select the relevant faculty member / supervisor from the available list to submit their research paper.
- Students will be able to check the status of their submission.
- Each submission will have a deadline, and students will not be able to submit the research paper if they miss the deadline.

FR-4: Research Paper Evaluation

- Faculty member/ Supervisor can evaluate the submitted papers based on predefined evaluation criteria (e.g., paper structure, clarity, originality, domain relevance, novelty and quality of research etc.).
- After evaluation of the research, the Faculty Members / Supervisors can update the status as “Accepted”, “Rejected”, “Needs Improvement”, “Accepted and Published”) and can add further comments/feedback against the research paper (if any).
- In case of rejection of a paper, a valid reason will be entered by the faculty member / supervisor accordingly.
- In case of status as Rejection or Needs Improvement, the faculty member / supervisor will be able to upload the commented file on the system.
- Only the research papers having the status “Accepted and Published” will be available on the system for public access.
- Faculty member / Supervisor can also view all the past evaluations and submissions for students.

FR-5: Document Updates and Versioning

- Students can update their research papers after submission if needed and the system will maintain the log of previous versions.

FR-6: Notifications and Alerts

- Notifications through email are sent to students upon submission, after evaluation, and when feedback is available.
- Faculty member / supervisor will be notified through email when new papers are submitted for evaluation.
- Admin will receive notifications through email regarding pending evaluations or issues.

FR-7: Search and Filter Mechanism

- For “accepted and published” papers, Admin, students, and faculty members / supervisors can search by keywords like, Student ID, Study Program Enrolled, Category, Date, and Author etc.

FR-8: Report Generation

- Admin can generate reports on submission statistics, evaluation progress, and list of published papers on the system.
- Faculty member / supervisor can generate reports based on evaluation of their respective students.

FR-9: Administrative Monitoring and Control

- Admin can manage user roles (add/edit/delete users).
- Admin can monitor submission status (accepted, pending, accepted, accepted and published), track evaluation progress and ensure timely feedback.

1.5.2 Non-Functional Requirements

NFR-1: Security and Data Protection

- Secure authentication and authorization mechanisms should be deployed.
- Storage of research papers should be encrypted to prevent unauthorized access.

NFR-2: Usability and User Experience

- System should be easy to use and navigate.
- Responsive design across various devices (desktop, mobile, tablet).

NFR-3: Performance and Scalability

- Fast loading time and efficient task execution.

- System should be capable of handling multiple users simultaneously.

NFR-4: Availability and Reliability

- System must always ensure uninterrupted access to data and query execution.
- Robust data backup and recovery mechanisms should be integrated.

NFR-5: Maintainability

- System should use modular components for easy updates and troubleshooting.
- Clear documentation and coding standards must be maintained to simplify future changes.

1.6 USE CASES AND USAGE SCENARIOS

1.6.1 USE CASE DIAGRAMS

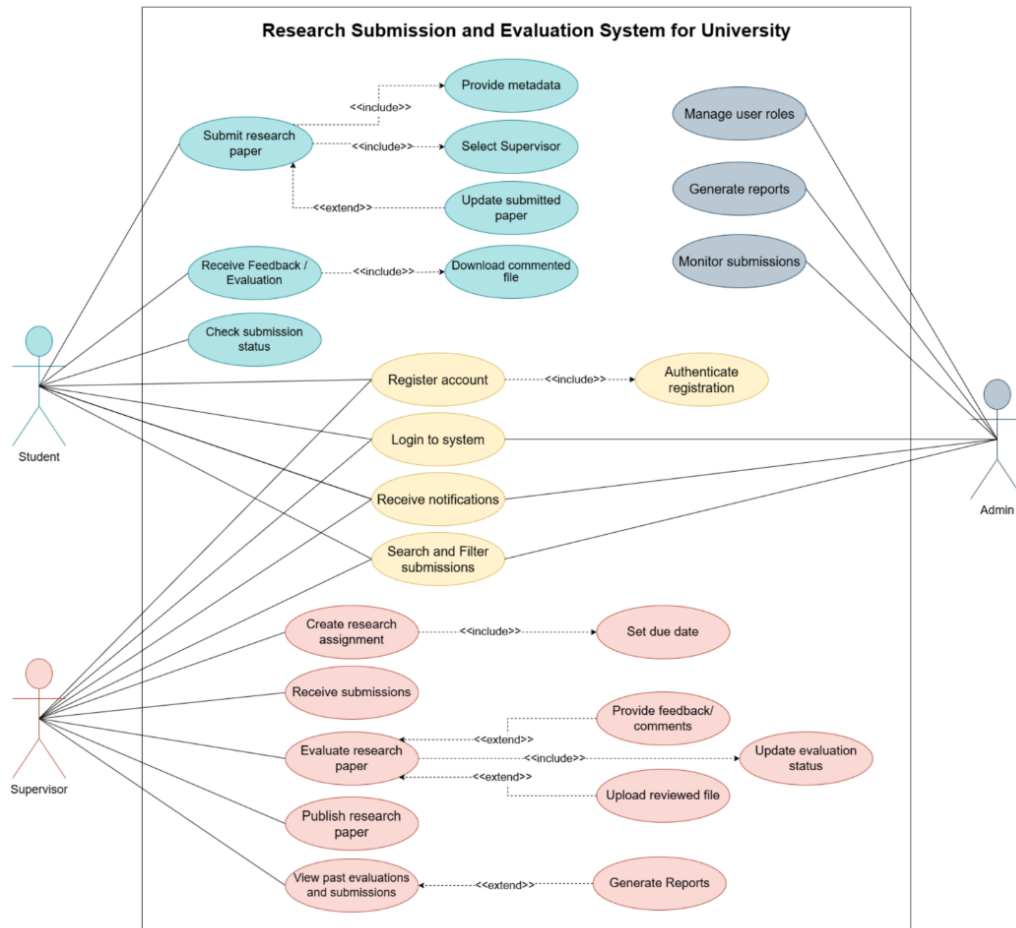


Fig: Use Case Diagram

1.6.2 Usage Scenarios

Use Case Title	Register Account
Use Case ID	UC-01
Actor(s)	Supervisors/Faculty members and Students
Description	System allows the students, supervisors/faculty members to register a new account through the website's login/register portal, by filling the registration form.
Pre-conditions	System must be active and connected to the database.
Actions	1. Students and supervisors will navigate to the main page of the website where they will find the 'login/register' option.

	- Once it's clicked, they will be presented with two options: login or register a new account. - The users will click the latter, and be presented with a registration form, which is necessary for account creation. - They will be asked to provide username and email address, institute name, set password, specify if they are student or supervisor. - They will hit the submit button once done.
Exceptions	Warning is triggered when some forms fields are left blank.
Post-conditions	- The admin will either accept or reject the account creation request based on the validity of user. - User account is created successfully once the registration is authenticated.
Alternative path	Students, supervisors/faculty members decide to cancel account registration.
Author	BC220203838

Use Case Title	Login to System
Use Case ID	UC-02
Actor(s)	Supervisors/ Faculty members, Students, Admin
Description	Users and admin will be able to login to their accounts through the website's login/register page by providing necessary credentials.
Pre-conditions	- System must be active and connected to live database. - Students and supervisors/ faculty members must have already registered accounts in the system.
Actions	- Admin, students, and supervisors will visit the main page of the website where they will find the 'login/register' option. - Once it's clicked, they will be presented with two options: login or register a new account. - They will choose "login". - They will be asked to provide credentials (username and password). - Once done, they will click on the "login" button.
Exceptions	- User account is not found in the database given that the provided credentials do not match existing accounts. - Incorrect username or password. - The website fails to connect to the database.
Post-conditions	Admin, students, and supervisors successfully login to their accounts and gain access the services associated with their roles.
Alternative path	- Users decide not to log into the system and navigate away from the login page. - Users choose to register for a new account instead of logging in.
Author	BC220203838

Use Case Title	Create Research Assignment
Use Case ID	UC-03
Actor(s)	Supervisors/ faculty members

Description	After the supervisor/ faculty member has prepared the research assignment file, they're able to upload it to the system and successfully post it for students to view and download.
Pre-conditions	• System must be active and available for use. • The supervisor must be logged into their account. • The assignment file must be in either PDF format or MS Word format.
Actions	1. The supervisor will navigate to the option "Upload research assignment" and click it. 2. The supervisor will be led to a portal and asked to choose the desired file from his/her device. 3. Once the file has been selected, the supervisor will click on the "upload" button. 4. Once the assignment has been loaded onto the system, the students will be able to find it in the "Assignments" section.
Exceptions	1. System fails to upload the assignment due network or server issues. 2. The file format is not supported or corrupted.
Post-conditions	• The assignment is uploaded successfully. • The students can access the file. • The supervisor receives a confirmation message that the file has been posted.
Alternative path	The supervisor decides to cancel uploading the assignment file.
Author	BC220203838

Use Case Title	Submit Research Paper
Use Case ID	UC-04
Actor(s)	Students
Description	The student is able to submit their research paper to the system, either in PDF format or MS Word format, for the supervisor to view and evaluate.
Pre-conditions	• System must be active and accessible. • Students must be logged into their account. • Students must provide meta data (student ID, study program, paper title, etc.) • Students must select the supervisor/ faculty member. • The assignment file must be in either PDF format or MS Word format. • Submission must only be made within the due date.
Actions	1. The student will navigate to the option titled as "Submit research assignment" and click it. 2. They will be ported to a portal that will ask them to choose the file from their device. 3. Once they've selected the file to upload, they will click on the "submit" button. 4. The option "Update research assignment" is displayed once the paper has been submitted, in case the student decides to submit an updated version of the file.
Exceptions	1. The system fails to upload the file due to network or server issues. 2. The file format is not supported or corrupted. 3. The student fails to submit the paper within the due date.
Post-conditions	• The file is uploaded successfully.

	• The supervisor can view the file and evaluate accordingly. • The student receives a confirmation message that the file has been submitted.
Alternative path	The student decides against uploading the research paper file and clicks on the “cancel” button.
Author	BC220203838

Use Case Title	Evaluate Research Paper
Use Case ID	UC-05
Actor(s)	Supervisors/Faculty members
Description	The supervisor/faculty member evaluates the submitted research paper based on predefined criteria. They can update the paper’s status, rate it through star rating and provide feedback or comments when necessary. If required, they can upload a commented version of the file for the student to view.
Pre-conditions	• System must be running and accessible. • Supervisor/ faculty member must be logged into their account. • A research paper must have been submitted by the student for evaluation.
Actions	1. The supervisor/ faculty member will navigate to the “Research paper evaluation” section. 2. They will select the submitted paper to evaluate. 3. The submitted paper will open in a separate page. 4. The supervisor/ faculty member will assess the paper based on criteria, including clarity, originality and relevance. 5. They will mark the paper with a status from the available options: “accepted”, “rejected”, “needs improvement”, “accepted and published”. 6. In case “rejected” or “needs improvement” is marked, a pop up will be displayed on the screen, asking to provide a valid reason or a comment, and if needed, upload a commented file. 7. The supervisor/ faculty member will be asked to rate the paper using star rating.
Exceptions	1. The system fails to load the submitted paper due to network or server issues. 2. They system fails to update the paper status due to network or server issues.
Post-conditions	• Supervisor receives a confirmation message “Research paper evaluated successfully”. • The research paper is successfully evaluated, and its status is updated. • Feedback or comments will be available for the student to view. • Paper marked as “accepted and published” becomes publicly accessible.
Alternative path	The supervisor/ faculty member decides not to evaluate the paper at the moment and exits the evaluation process, leaving the paper unevaluated.
Author	BC220203838

Use Case Title	Receive Feedback/ Evaluation
Use Case ID	UC-06
Actor(s)	Students

Description	The student receives evaluation from the supervisor, which may include feedback, rating, evaluation status, a commented file, rejection reason, or a notification in case the paper is accepted and published.
Pre-conditions	1. System must be up-to-date, active and available for use. 2. Supervisor/ faculty member must have evaluated the paper successfully. 3. Student must be logged into their account.
Actions	1. Student navigates to the “research paper submission” page, where a table is displayed. 2. Feedback from the supervisor, including rating and comments, will be displayed in a dedicated column. 3. Student clicks the "commented file” option and download the file containing the supervisor’s remarks.
Exceptions	1. System fails to load feedback due to network or server issues. 2. The commented file is corrupted.
Post-conditions	• Student successfully views the feedback and evaluation. • The student downloads the commented file if available.
Alternative path	The student views only the evaluation status and rating but does not download the commented file (if any).
Author	BC220203838

Use Case Title	Check Submission Status
Use Case ID	UC-07
Actor(s)	Students
Description	Students can determine the status of their research paper through the status updated by the supervisor, as Pending, Accepted, Rejected, Needs Improvement, Accepted and Published.
Pre-conditions	1. Student must be logged into their account. 2. Student must have submitted the research paper.
Actions	1. Student navigates to the “Check Submission Status” page. 2. The system displays list of submitted papers along with the status among the following: Pending – paper is awaiting evaluation Accepted – paper has been approved Rejected – paper does not meet the criteria Needs improvement – paper needs few enhancements Accepted and Published – paper has been accepted and published.
Exceptions	1. Status could not be retrieved due to network or server issues. 2. Status not available because the student has not submitted the paper.
Post-conditions	• Student successfully views the submission status. • If necessary, student takes corrective action after reviewing feedback.
Alternative path	None
Author	BC220203838

Use Case Title	Monitor Submissions
Use Case ID	UC-08
Actor(s)	Admin
Description	System enables the admin to track submission status of research papers, providing updates on whether a paper is accepted, rejected, pending, needs improvement, or has been accepted and published.
Pre-conditions	• System must be online and available for use. • Database must be connected and accessible. • Research papers must have been submitted by students for evaluation.
Actions	1. Admin navigates to “Monitor Submissions” section on the admin panel. 2. A list of submitted research papers is displayed, where each entry is labeled with a status indicator (pending, accepted, rejected, needs improvement, accepted and published). 3. Admin views submission details, including timestamps, authors, and evaluation history. 4. Admin notifies relevant users if any submission requires attention or has been marked for publication.
Post-conditions	• Submission status is successfully monitored and recorded. • System maintains an accurate log of submission activities for future references.
Exceptions	1. System fails to retrieve submission information due to network or server issues. 2. Research paper does not have a status due to incomplete evaluation process.
Alternative path	• Admin decides to postpone monitoring process and checks submission status at a later time. • Admin filters submissions to focus on specific statuses instead of viewing all the submissions.
Author	BC220203838

Use Case Title	Manage User Roles
Use Case ID	UC-09
Actor(s)	Admin
Description	System assists admin in efficiently managing user roles by adding, editing, deleting, authorizing users. The admin has full control over user access and permissions within the system.
Pre-conditions	1. System must be active and accessible. 2. Admin must be logged into their account.
Actions	1. In the admin panel, the admin selects “Manage Users” option from the navigation menu. 2. Once clicked, a structured table displaying all users appears, listing their roles and statuses. 3. Admin is presented with multiple options: • Add User: Creates a new user profile with assigned roles. • Edit User Information: Modifies details such as name, email, role etc. upon the user’s request. • Delete User: Removes user profile from the system permanently.

	• Authorize New User: Grants necessary permissions to new user, after accepting/rejecting their registration request. 4. Changes are validated and applied accordingly, ensuring a secure user role management process.
Post-conditions	Users are successfully added, updated, authorized, or removed.
Exceptions	1. System fails to respond to requests due to network or server issues. 2. Admin's login credentials are incorrect.
Alternative path	The admin decides not to make any modifications and exits the "Manage Users" panel.
Author	BC220203838

Use Case Title	Receive Notifications
Use Case ID	UC-10
Actor(s)	Admin, Students, Supervisors/ Faculty members
Description	The system ensures that users receive notifications based on their roles. Admins are informed about system issues and pending evaluations, students receive updates on their submission progress, while supervisors/faculty members are notified about new submissions and inquiries by students.
Pre-conditions	• System must be online and operating correctly. • Database must be connected. • Users must have valid email addresses registered in the system.
Actions	1. The system generates notifications based on certain triggers and user interactions. 2. Admin will receive email alerts regarding system performance issues and pending evaluations. 3. Students will receive emails confirming the submission of their research paper, updates on evaluation status or feedback from supervisor, and notifications if the paper is accepted and published. 4. Supervisors/ faculty members will receive emails notifying them of new research paper submissions by students and direct inquiries from students regarding the assignment and feedback.
Exceptions	1. Email delivery fails due to system or network issues. 2. User's registered email is incorrect or inactive.
Post-conditions	• Notifications are successfully sent and received by the appropriate users. • Users are informed about relevant updates in the system.
Alternative path	Users forget to check their emails.
Author	BC220203838

Use Case Title	Search and Filter Submissions
Use Case ID	UC-11

Actor(s)	Admin, students, supervisors/faculty members.
Description	System allows users to view and access the published research papers through a search and filtering mechanism.
Pre-conditions	• System must be active, accessible, and up to date. • Users must be logged in to their respective accounts.
Actions	1. Users navigate to the portal labeled as “Published papers” on the website. 2. They click the title and are redirected to a page displaying a list of all the papers that have been accepted and published. 3. Users can view a paper by clicking its title. 4. To access certain papers, users can utilize the search and filter mechanism. 5. A search bar is positioned at the top of the page along with several filter options, including: • Paper Title • Student ID • Study Program Enrolled • Category • Date • Author 6. Users select a filter criteria and enter the relevant keywords. 7. The system retrieves and displays the papers that match the specified criteria or displays “paper does not exist” in case the keywords do not match.
Exceptions	• System faces network or server issues. • Search results do not exist in the system database.
Post-conditions	Users successfully access the published research paper of their choice.
Alternative path	Users choose to browse papers manually instead of using search and filter option. Users decided not to view any research paper and navigate back to the main page.
Author	BC220203838

Use Case Title	Generate Reports
Use Case ID	UC-12
Actor(s)	Admin, Supervisors/faculty members
Description	System enables admin to generate reports regarding submission statistics, evaluation progress, and the list of published papers. While the supervisors/faculty members can generate reports based on the evaluation of their students.
Pre-conditions	• System must be active and accessible while database remains connected. • Respective users must be logged into their accounts to gain access to this service. • Relevant data must be available for the report to be generated.
Actions	1. Admin or supervisor/faculty member navigates to the “Generate Report” section of the system. 2. Admin selects the type of report they wish to generate, from the following: • Submission statistics • Evaluation progress • Published papers 3. Supervisor/faculty member selects the option to generate report based on student evaluations.

	4. The system retrieves respective data based on the specified parameters. 5. Once the users click on the “Confirm report generation” button, a report is created and displayed in a separate page. 6. Users can either export or print the report for documentation purposes.
Exceptions	1. System fails to connect to the database, preventing report generation. 2. Missing data results in an incomplete or inaccurate report.
Post-conditions	• The report is generated successfully and accessible to the user. • User can choose to download or print the report for future use. • Any necessary modifications can be made before finalizing the report.
Alternative path	• User cancels the report generation process and navigates back to main page. • User decides to select a different report type instead.
Author	BC220203838

1.7 DEVELOPMENT METHODOLOGY

1.7.1 Chosen Methodology

The VU process model is the adopted methodology for this project.

VU Process Model:

It is a combination of Waterfall and Spiral models. In other words, it can be referred to as a hybrid approach to the software development process. By combining the advantageous features of both Waterfall and Spiral models, it becomes an efficient and well-structured model to guide the development process. It adopts the phases of waterfall model and integrates the risk-assessment feature of the spiral model, into each stage to ensure that the process goes through needed refinements and gather proper feedback from the supervisor/ user. The project’s process will not progress to the next step until the previous step is confirmed to meet the requirements and is free of errors.

1.7.2 Reasons for Chosen Methodology

Reasons why this methodology is suitable for the project:

- It combines the strengths of both the Waterfall and Spiral models.
- It provides a clear, sequential, and straightforward method that simplifies project management.
- It prioritizes risk assessment at each stage to smoothen the identification and mitigation of risks and make the development process more reliable.
- It ensures that each step is refined, improved, and aligns with the requirements.
- It integrates regular feedback from the supervisor/user who ensures that the system is on track with the expectations and requirements.
- Through iterative refinements, it reduces the likelihood of major revisions at the later stages.

Diagram for VU process model is shown below.

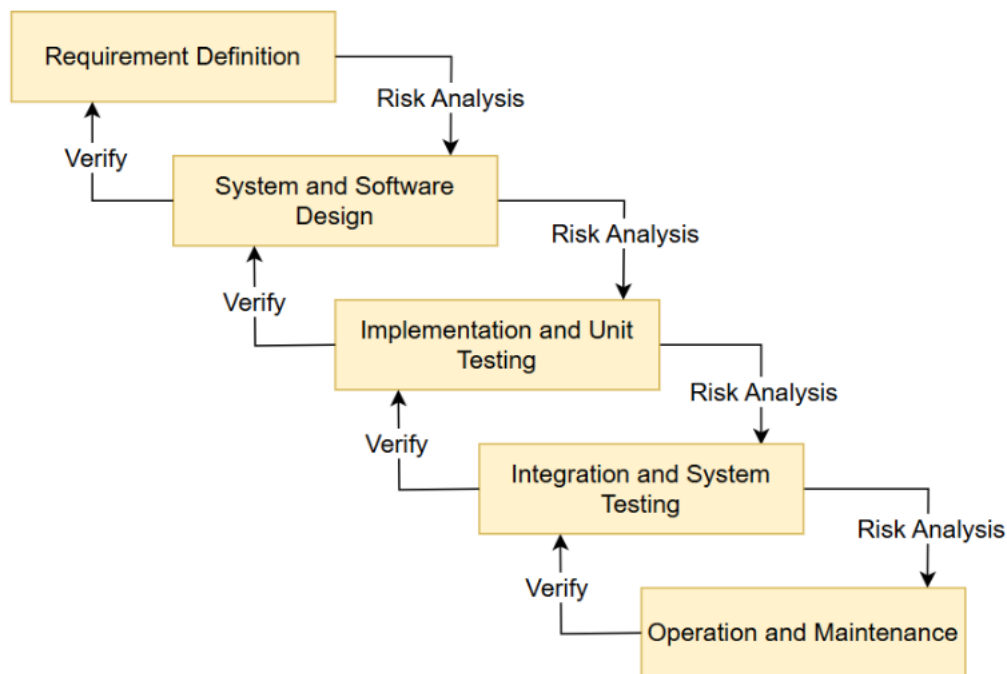


Fig: VU Process Model Diagram

1.7.2 Work Plan (Gantt Chart)

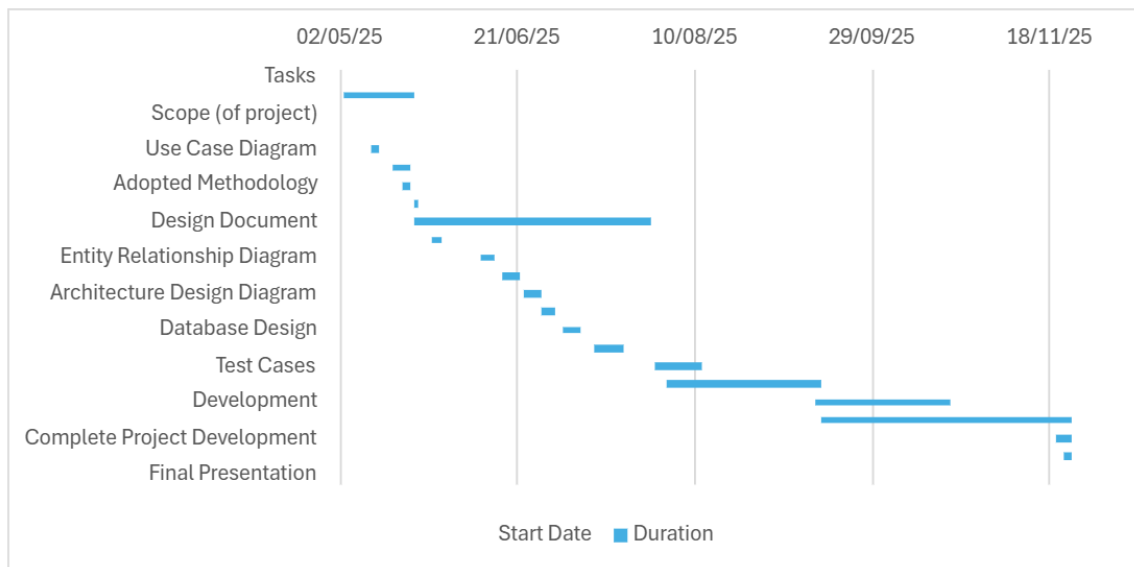


Fig: Work Plan (Gantt Chart)

1.7.2 Project Schedule (Submission Calendar)

Sr. No.	Assignment Title	Start Date	End Date
1	SRS Document	02 May 2025	22 May 2025
2	Design Document	23 May 2025	29 Jul 2025
3	Prototype Phase	30 Jul 2025	12 Sep 2025
4	Final Deliverable	13 Sep 2025	24 Nov 2025

CHAPTER 2

Designing the Project

2.1 INTRODUCTION

This chapter serves as the design document, a comprehensive piece of documentation that defines how the system will be structured, designed, implemented, and operated. It provides a clear blueprint for developers, ensuring that the software development process remains consistent, well-organized, and free from ambiguity.

2.2 PURPOSE

The purpose of the design document is to define the system architecture, outline how various components interact, reduce development risks, and prevent misunderstandings by clarifying requirements and expectations.

It also improves maintainability by simplifying future updates and debugging, while streamlining implementation through a clear roadmap that ensures efficiency and a well-built system.

2.3 SCOPE

The design document structures the entire system in a clear and simpler format. It encompasses the following,

1. **Entity Relationship Diagram:** An ERD visualizes the structure of the database by showing how different entities relate to each other. It helps in designing and understanding the database systems by showing the relationship between the entities and lists their attributes.
2. **Sequence Diagram:** A sequence diagram is used to illustrate the interactions between objects or components in the system over time. It visualizes the order and the timing in which the interactions or events occur between different objects in the system. It is also called an 'Event Diagram'.
3. **Architecture Design Diagram:** This diagram serves as a high-level display of the system's structure. It provides a tiered architecture that outlines the system's layers including the user interface, the business logic and database. It guides the implementation process by ensuring a smooth integration between the different modules.
4. **Class Diagram:** A class diagram represents the structure of the system by defining classes, their attributes, functions and the relationships connecting them. It simplifies the process of understanding how the components of the system interact.
5. **Database Design:** A database design diagram showcases how the data relevant to the system is stored and managed within the database. It helps visualize how the information will reside in the database.

6. **Interface Design:** Interface design helps in visualizing the system to be developed. It highlights the layout, the UI, and other prominent features of the software/system.

2.4 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

Definitions

- **Design Document** – A comprehensive guide that outlines how the system will be structured, designed, implemented, and function. It ensures clarity, consistency, and reduces ambiguity in the development process.
- **Entity Relationship Diagram (ERD)** – A diagram that visualizes the structure of the database by showing entities, attributes, and relationships.
- **Sequence Diagram** – A diagram that illustrates the interactions between objects or components in the system over time, also known as an *Event Diagram*.
- **Architecture Design Diagram** – A high-level representation of the system's structure, showing layers such as user interface, business logic, and database.
- **Class Diagram** – A diagram that defines classes, their attributes, functions, and relationships, simplifying the understanding of system components.
- **Database Design** – The structured representation of how data is stored, organized, and managed within the system's database.
- **Interface Design** – The visualization of the system's layout, user interface (UI), and prominent features to ensure usability.
- **Crow's Foot Notation** – A method used in ERDs to represent cardinality (the number of relationships between entities).

2.5 ARCHITECTURAL REPRESENTATION (ARCHITECTURE DIAGRAM)

Tiered Architecture of the System

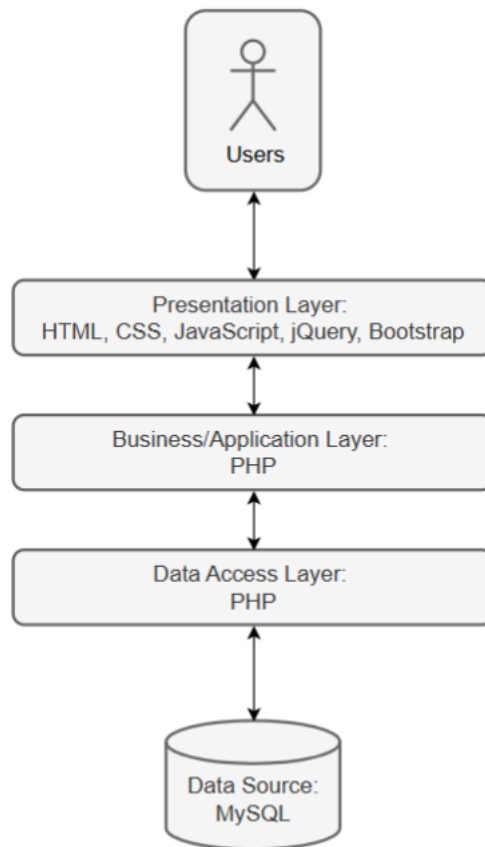


Fig: Architecture Design Diagram

2.6 DYNAMIC MODEL: SEQUENCE DIAGRAMS

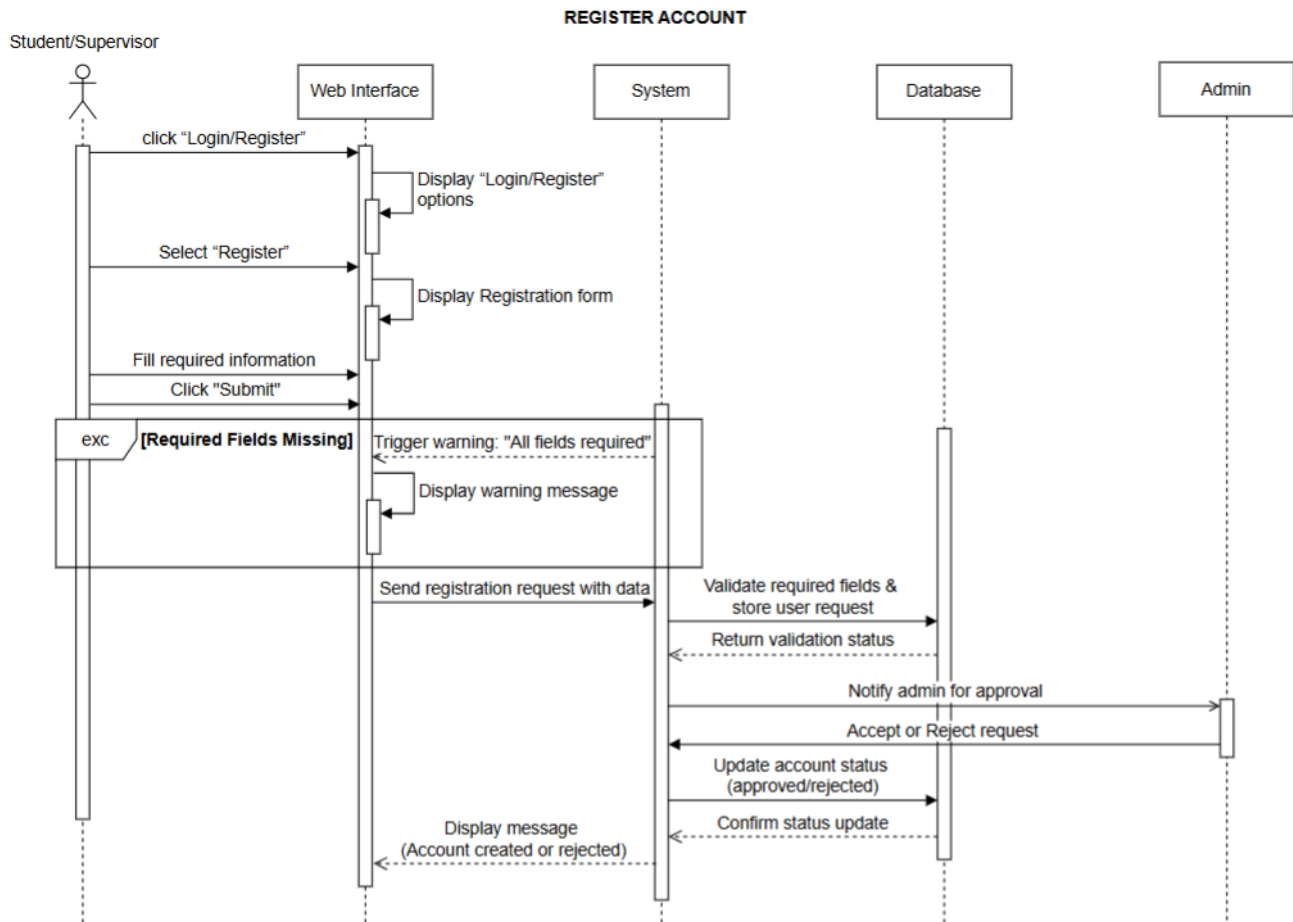


Fig: sequence diagram-use case-01-Register Account

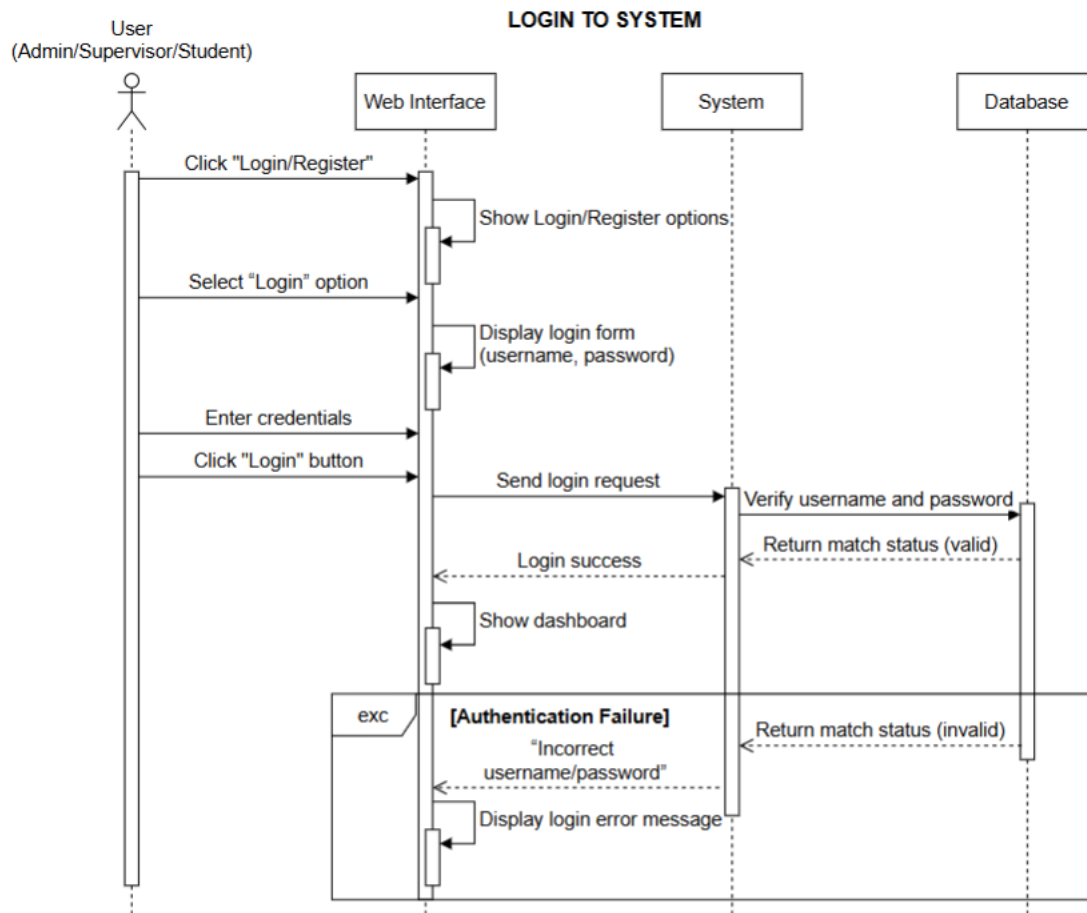


Fig: sequence diagram-use case-02-Login to System

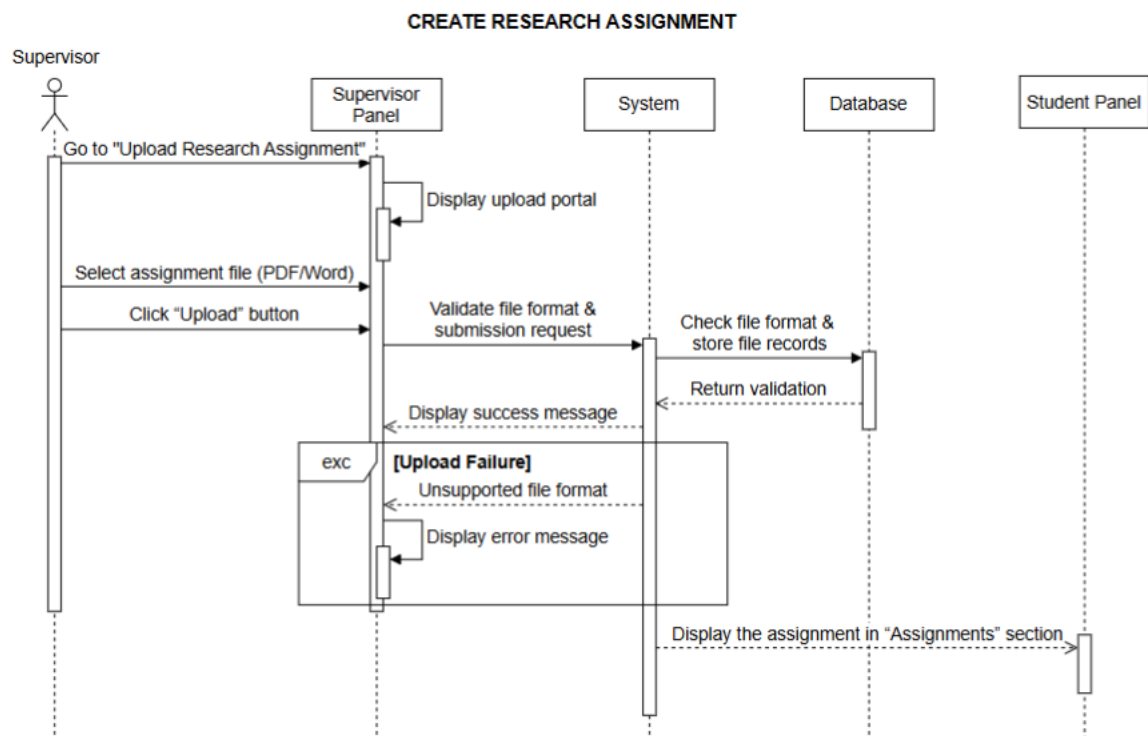


Fig: sequence diagram-use case-03-Create Research Assignment

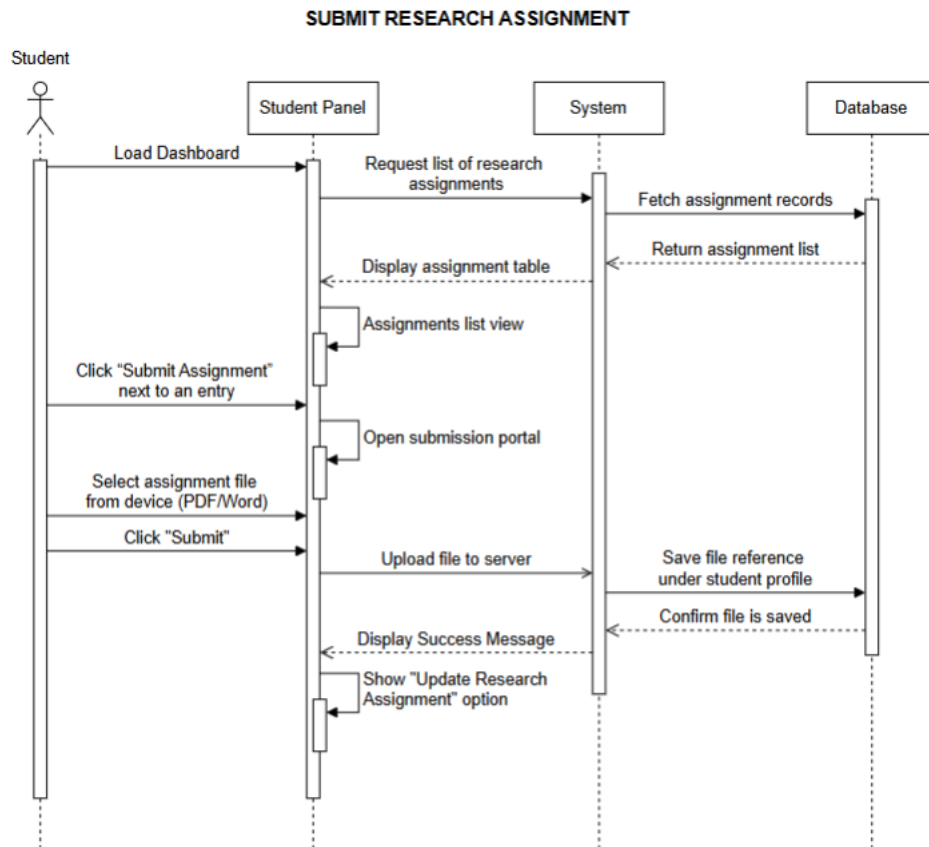


Fig: sequence diagram-use case-04-Submit Research Assignment

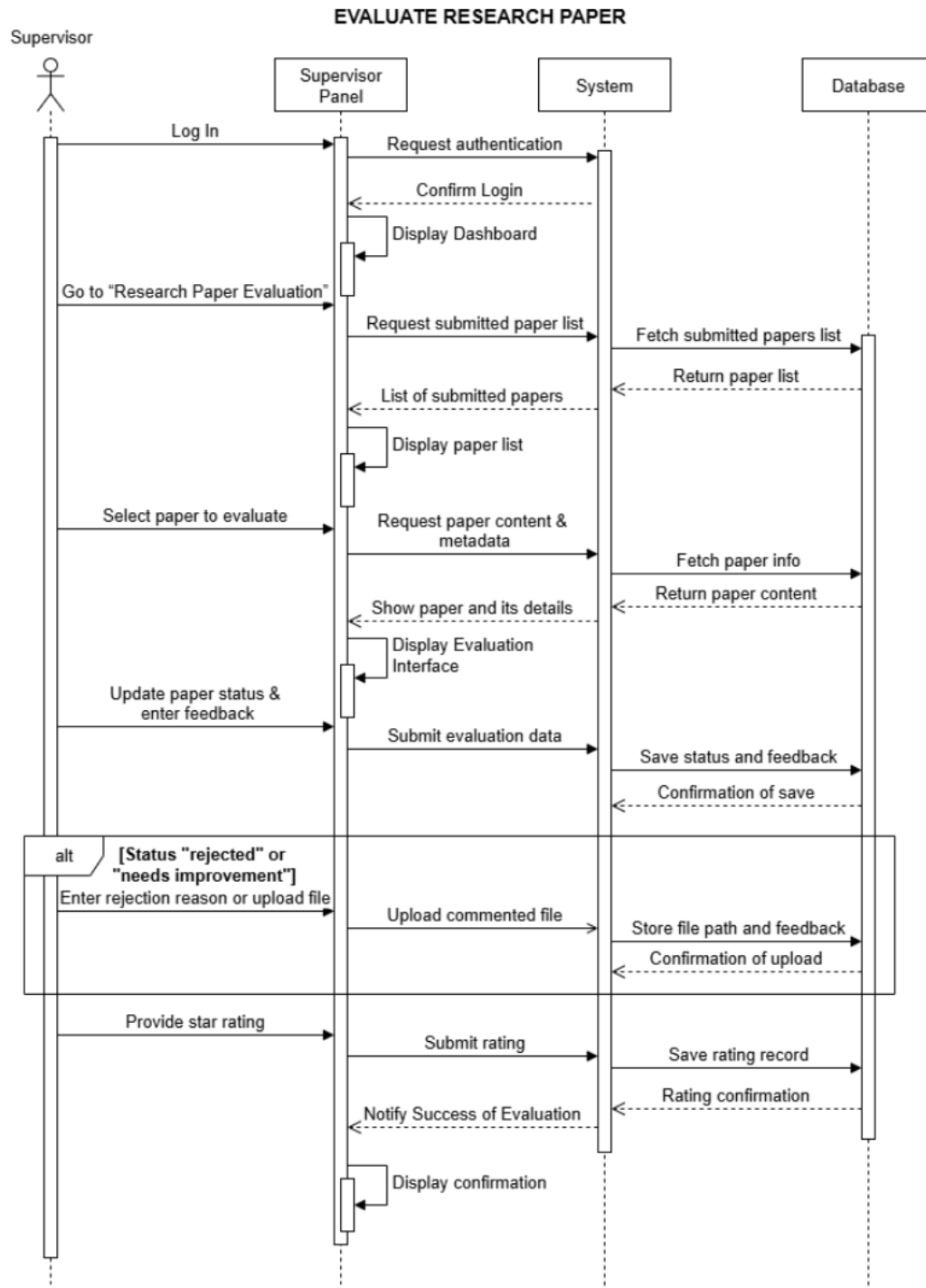


Fig: sequence diagram-use case-05-Evaluate Research Paper

RECEIVE FEEDBACK/ EVALUATION

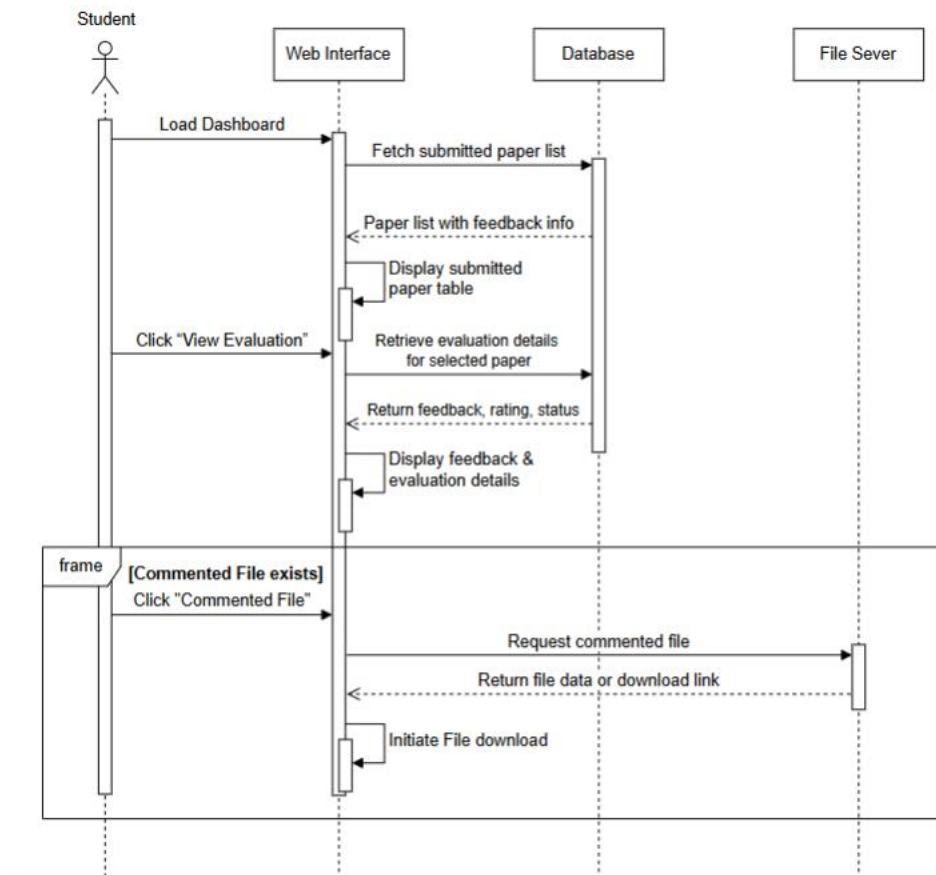


Fig: sequence diagram-use case-06-Receive Feedback

CHECK SUBMISSION STATUS

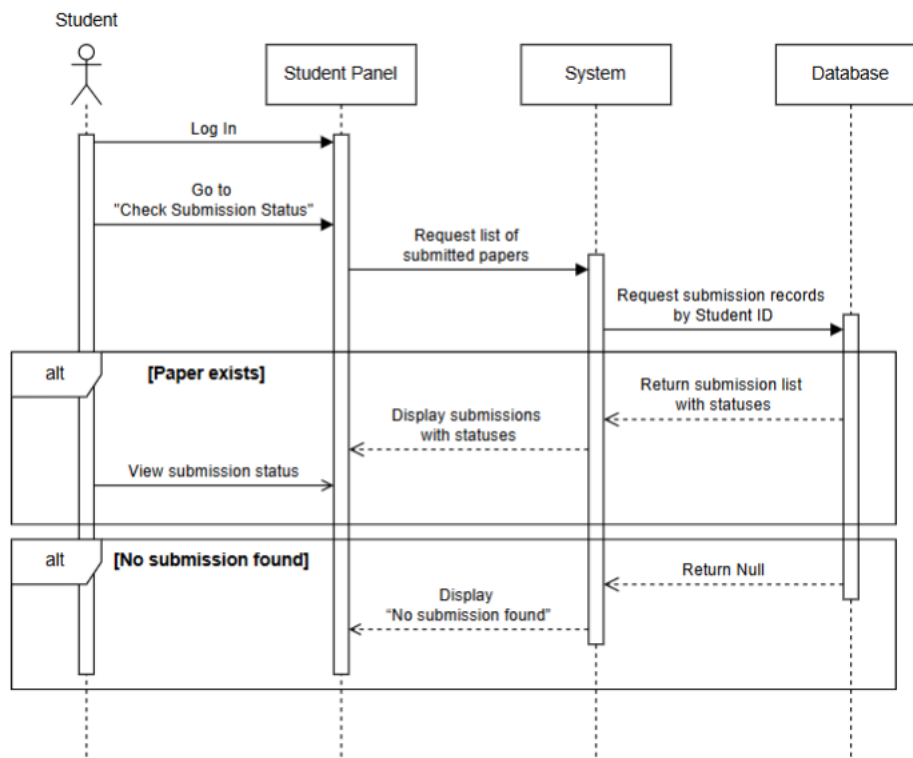


Fig: sequence diagram-use case-07-Check Submission Status

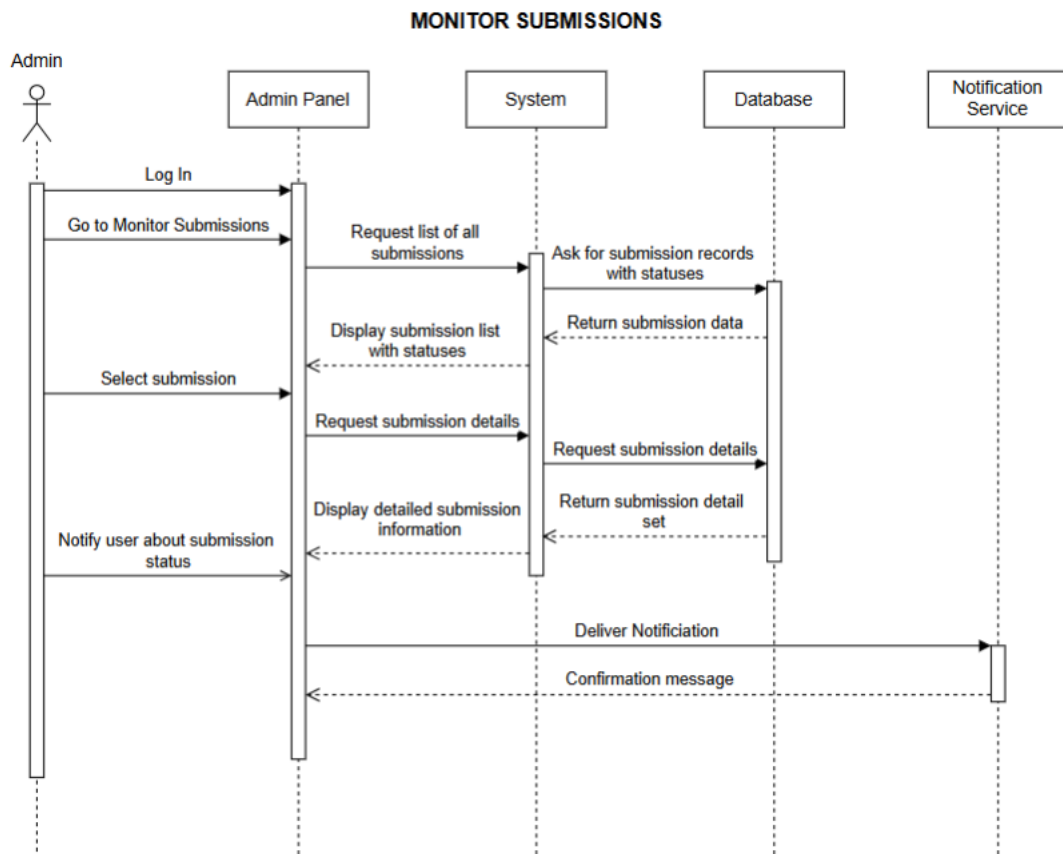


Fig: sequence diagram-use case-08-Monitor Submissions

MANAGE USER ROLES

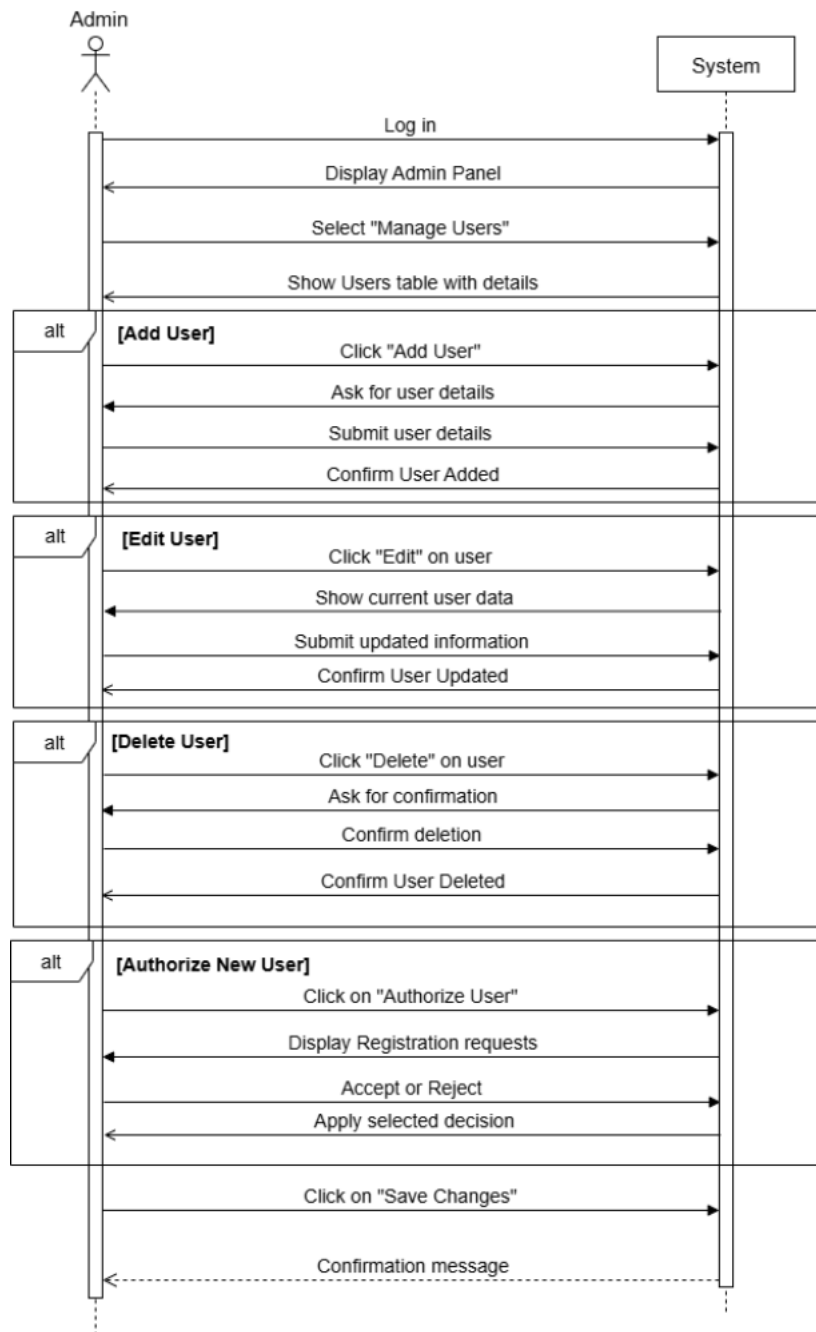


Fig: sequence diagram-use case-09-Manage User Roles

RECEIVE NOTIFICATIONS

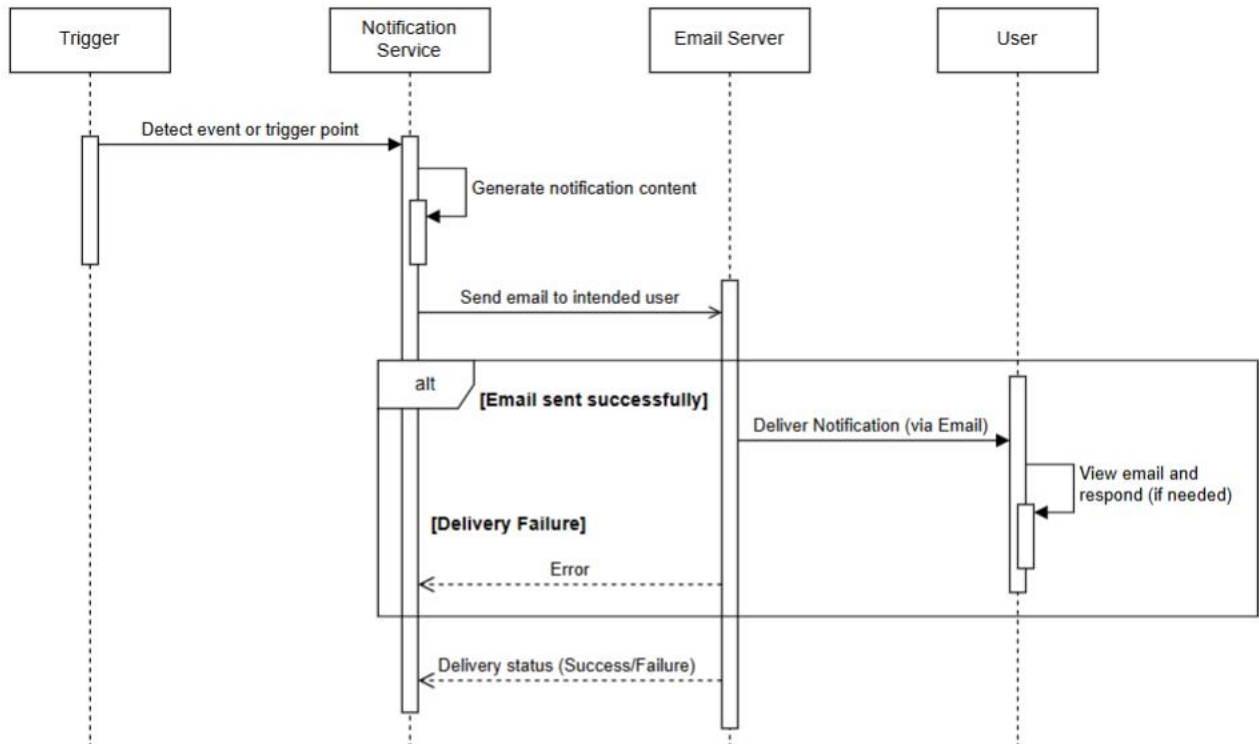


Fig: sequence diagram-use case-10-Receive Notifications

SEARCH AND FILTER SUBMISSIONS

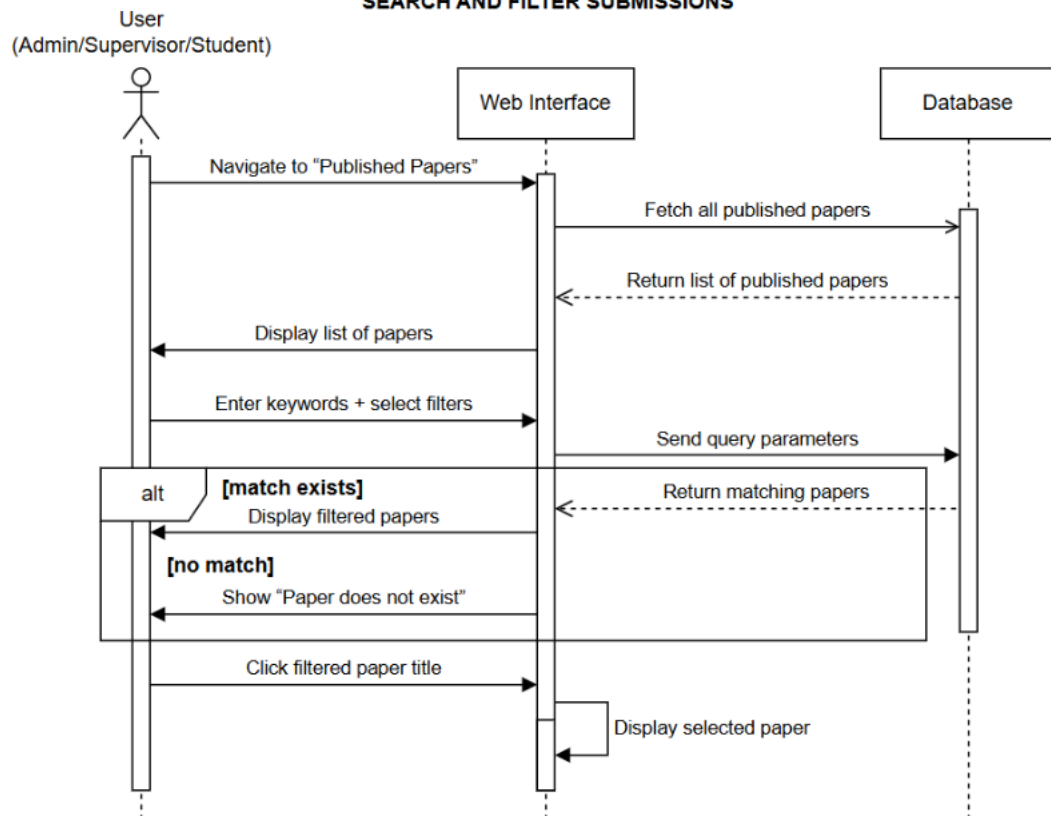


Fig: sequence diagram-use case-11-Search and Filter Submissions

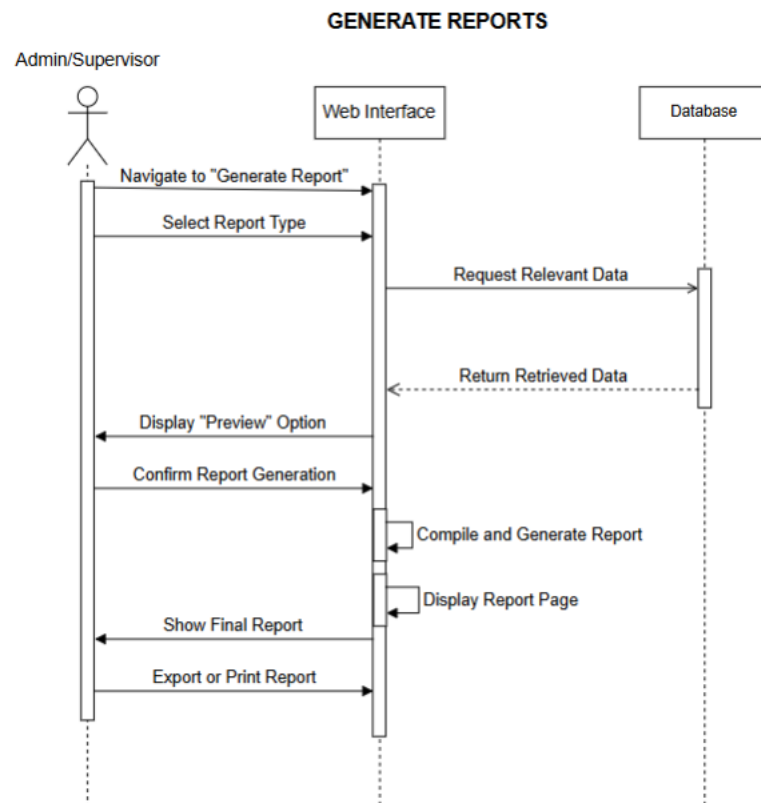


Fig: sequence diagram-use case-12-Generate Reports

2.7 OBJECT MODEL/LOGICAL MODEL: CLASS DIAGRAM

Class Diagram

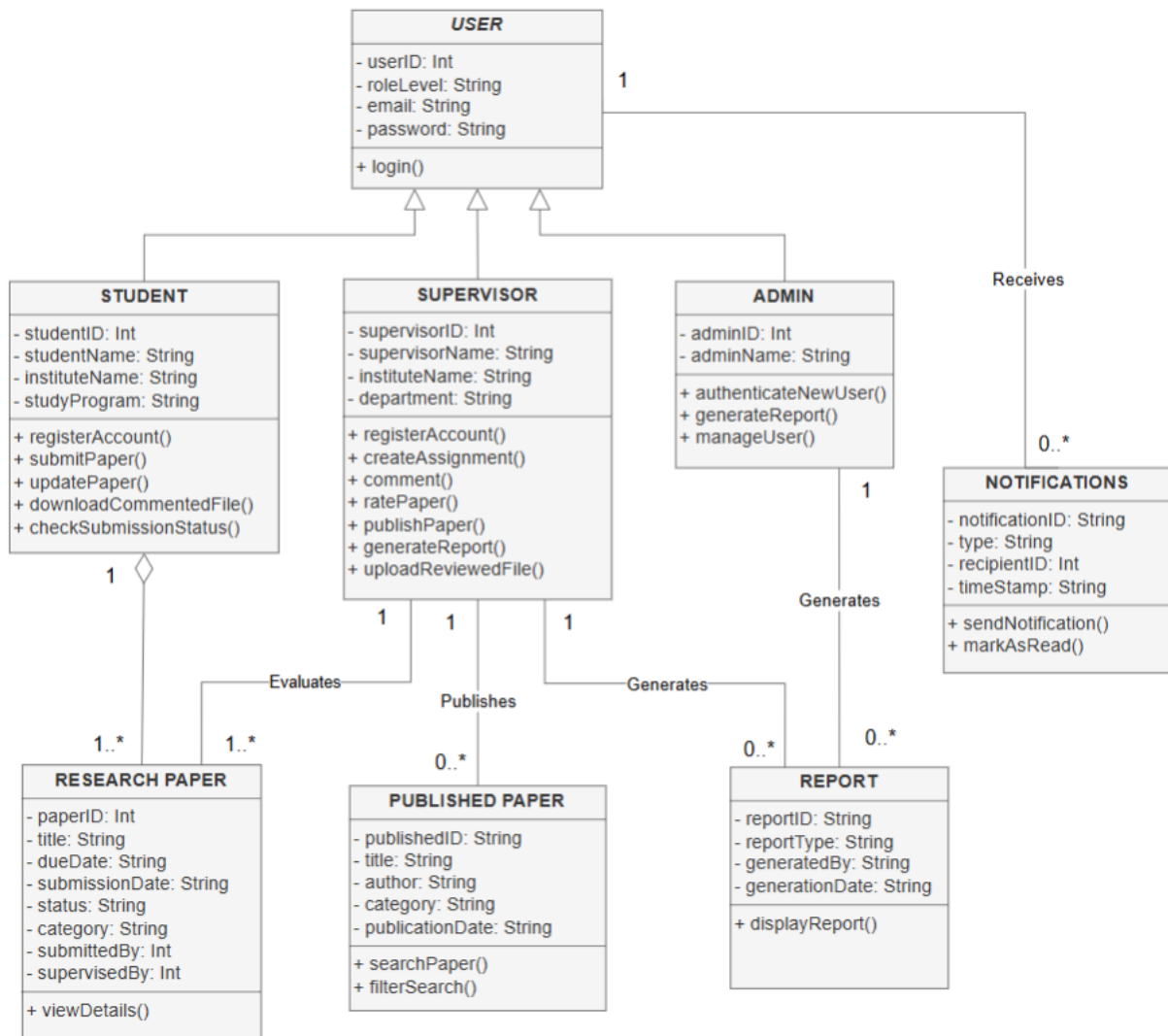


Fig: Class Diagram

2.8 DATABASE MODEL (DATABASE DIAGRAM)

Database Design Diagram

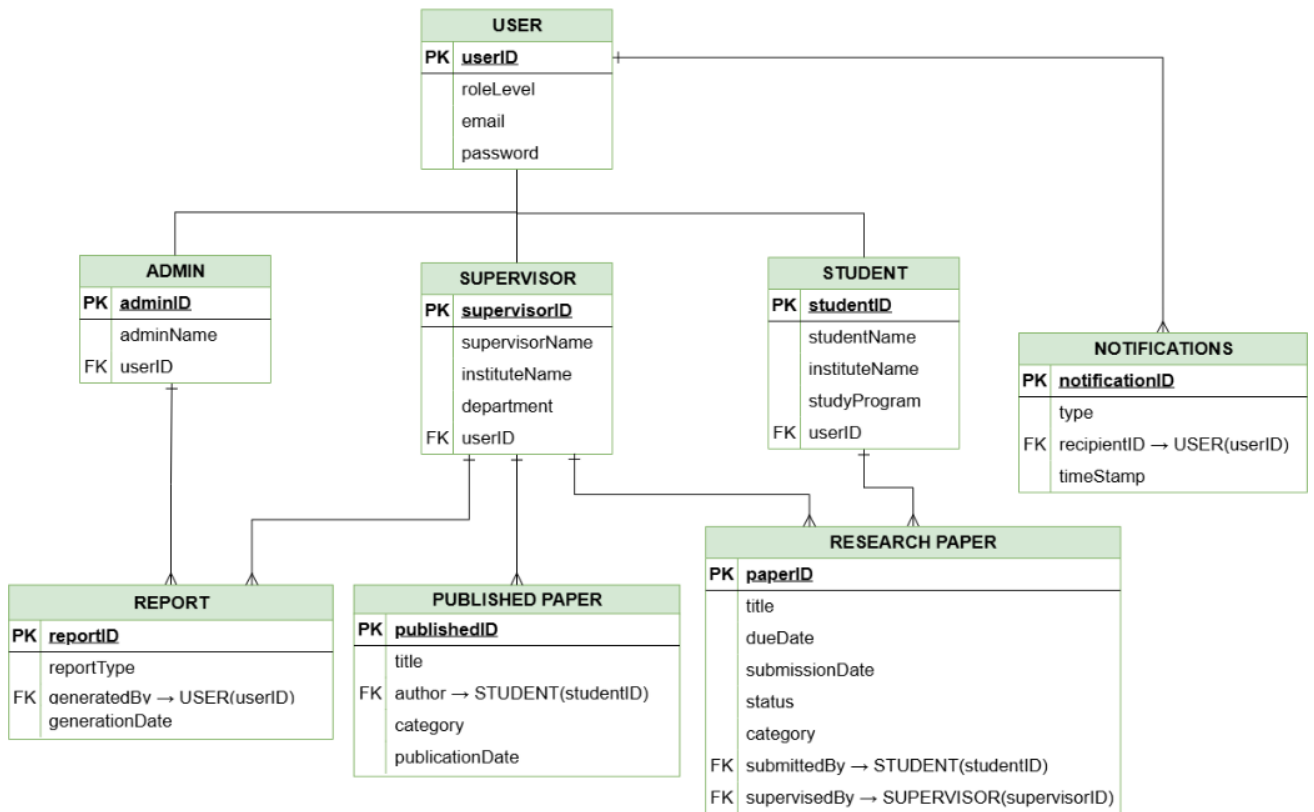


Fig: Database Design Diagram

2.9 GRAPHICAL USER INTERFACES

1. Landing Page

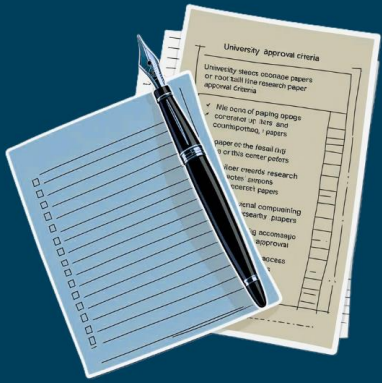


[HOME](#)
[PUBLISHED PAPERS](#)
[SUPERVISOR LIST](#)
[LOGIN/SIGN UP](#)

RESEARCH SUBMISSION AND EVALUATION SYSTEM

Welcome to the web-based project developed for universities to facilitate and manage the submission, evaluation, and feedback process of research papers. This platform is designed to streamline the handling of research papers, and enhance communication between students and supervisors.

[Read More](#)



2. Sign Up Page



SIGN UP

Get started with streamlined research submission and evaluation for students and supervisors.

You are a

STUDENT

SUPERVISOR

Full Name

Program

Email

Year of Study

Set Password

Institute Name

Confirm Password

SUBMIT

CLEAR ALL

3. Student Interface



PROFILE

HOME

EVALUATION STATUS

ASSIGNMENT DETAILS

HELP

LOGOUT



Welcome Student!

NOTIFICATIONS

No new notifications.

[VIEW AVAILABLE ASSIGNMENTS](#)

[VIEW EVALUATION STATUS](#)

ASSIGNMENTS

NO.	TITLE	START DATE	DUE DATE	SUBMIT	STATUS
-----	-------	------------	----------	--------	--------

4. Repository for Published Papers



HOME PUBLISHED PAPERS SUPERVISOR LIST [LOGIN/SIGN UP](#)

PUBLISHED PAPERS

SEARCH

TITLE	AUTHOR	STUDENT ID	STUDY PROGRAM	CATEGORY	PUBLISHED DATE

5. Admin Interface



PROFILE

HOME

MONITOR SUBMISSIONS

MANAGE USERS

GENERATE REPORTS

LOGOUT



Welcome Admin!

System Notifications

• System Issue Detected

Test issue: simulated system problem for debugging.

23 Nov 2025 15:41

System Overview

Total Users: 12

Students: 5

Supervisors: 6

Pending Requests: 0

Recent Registrations

Supervisor (supervisor)

Student (student)

CHAPTER 3

Development

3.1 DEVELOPMENT PLAN (ARCHITECTURE DIAGRAM)

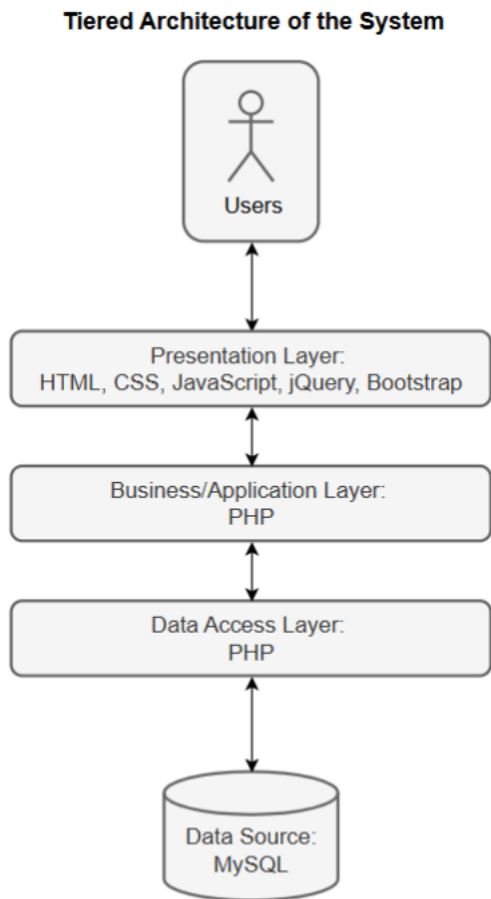


Fig: Architecture Design Database

3.2 Tools And Technologies Used

Tool	Purpose
HTML	To provide the structural foundation of web pages, defining their content and layout.
CSS	To provide styling and formatting for web pages, ensuring consistent design, colors, and layout.
JavaScript	To add interactivity and dynamic behavior, enabling features such as validation, animations, and real-time updates.
PHP	To handle server-side logic, process requests, and connect the application to databases for dynamic content.
XAMPP	To offer a local development environment by bundling Apache, MySQL, PHP for testing before deployment.
MySQL	To manage structured data storage in a relational database, supporting queries and secure retrieval.
jQuery	To simplify JavaScript programming with functions for DOM manipulation, event handling, and AJAX requests.
Visual Studio Code	To serve as a lightweight yet powerful code editor with debugging, extensions, and version control integration.

MailTrap.io	To create a safe environment for testing email functionality, capturing outgoing messages without sending them to real users.
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REFERENCES

YouTube

Gantt Chart — [Gantt Chart Excel Tutorial](#)

Sign-up/Login — [Signup and Login with PHP and MySQL](#)

Websites

CSS Responsive — [Responsive Web Design Media Queries](#)

SQL Joins — [SQL Joins](#)

Handling Errors with PHP — [PHP: Basics - Manual](#)

Layout Design Inspiration — [Pinterest.com](#)

APPENDIX

Chapters	Title	Pages
1	Gathering and Analyzing Info (Requirement Gathering Phase)	9 - 22
2	Designing the Document (Design Phase)	23 - 40
3	Development (Architecture Phase)	41 - 43

As we reach the end of this report, thank you for going through my work.

The completion of this project reflects countless hours of effort, research, and refinement. It has been a rewarding experience that strengthened my technical expertise and problem-solving mindset. More importantly, it has shown me that persistence is the key to turning challenges into achievements.